CSIRO PUBLISHING

Marine and Freshwater Research, 2018, **69**, 473–476
https://doi.org/10.1071/MF17189

Short Communication

Low-cost device for retrieval of moorings deployed with underwater remote release systems

D. A. Crook[A,C], *B. J. Adair*[A] *and L. J. Hetherington*[A,B]

[A]Research Institute for the Environment and Livelihoods, Charles Darwin University, Darwin, NT 0909, Australia.
[B]Present address: Nature Glenelg Trust, PO Box 2177, Mount Gambier, SA 5290, Australia.
[C]Corresponding author: Email: david.crook@cdu.edu.au

Abstract. Underwater acoustic and timed-release systems are commonly deployed with sensors and other equipment for ecological and oceanographic research in aquatic environments. These systems allow equipment deployed at depth to be remotely released from their moorings for retrieval. However, a current limitation is that most systems do not incorporate mechanisms for retrieval of the mooring anchor and attachments (e.g. shackles, cable). Herein we provide a detailed description and evaluation of a 'retrieval pod' that can be attached directly to remotely released underwater devices to facilitate mooring retrieval. The system was tested using acoustic release telemetry receivers under field conditions in a large estuary in the Northern Territory, Australia. Six receivers fitted with pods were retrieved from the field after two consecutive 6-month deployments (12 retrievals in total) with no failed mooring retrievals. Although our tests were confined to shallow depths (<8 m), it should be straightforward to modify the system for use in deeper water by increasing the spool capacity. The retrieval pods are reusable, require minimal construction skills and can be assembled from basic equipment available at most hardware or boating stores, making them a convenient and low-cost option for the retrieval of moorings deployed with remote release systems.

Additional keywords: acoustic telemetry, fish migration, mooring retrieval, oceanography, transceiver.

Received 1 March 2017, accepted 30 August 2017, published online 21 September 2017

Introduction

The remote deployment of sensors and other equipment for collecting *in situ* physical, biological and behavioural data is an increasingly used approach for oceanographic and ecological research in aquatic environments (Hussey *et al.* 2015). For example, studies of the movements of aquatic animals commonly deploy fixed arrays of acoustic telemetry receivers that are strategically located to provide specific information on the movements of tagged animals and are retrieved periodically for data download (Heupel *et al.* 2006). Receivers in these arrays are often deployed by suspending them between an anchor and a subsurface float, and retrieval involves grappling, diving or the use of remotely operated vehicles (Stark *et al.* 2005; Heupel *et al.* 2006; Webber 2009).

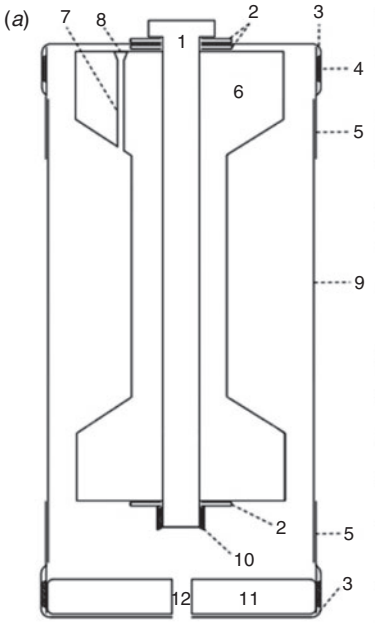
Acoustic and timed-release systems are also commonly deployed in acoustic telemetry and oceanographic studies, with sensors remotely released from their moorings using a mobile transceiver and hydrophone or via predictable corrosion of a metallic release or other timed mechanism. Once released, the equipment floats to the surface under the buoyancy of a subsurface float for retrieval (Heupel *et al.* 2006; Huveneers *et al.* 2016). Although the availability of remote release systems overcomes the need to locate and retrieve equipment with time-consuming and costly methods such as grappling, diving or remotely operated vehicles, a current limitation is that most systems do not incorporate mechanisms for the retrieval of the mooring weight and attachments (e.g. shackles, cable) with which the devices are anchored during deployment. The cost of replacing moorings and associated attachments upon each data download, as well as the potentially adverse environmental effects and public perception issues of discarding moorings, may reduce the utility of this type of equipment in many instances.

Although systems for retrieving moorings from acoustic release devices using containers with coiled rope have been used in several previous studies (e.g. Fujioka *et al.* 2010; Huveneers *et al.* 2016), to our knowledge detailed technical descriptions and evaluations of the performance of such systems are not currently available. This note describes a simple, low-cost device constructed from basic hardware that can be attached directly to underwater remote release systems to facilitate the retrieval of receiver moorings.

Retrieval pod

The retrieval pod described herein was developed using VR2AR acoustic release receivers (AMIRIX Systems, Vemco Division, Halifax, NS, Canada; https://vemco.com/products/vr2ar-acoustic-release-and-transceiver/, accessed 13 September 2017). The VR2AR features a release lug that can be activated using a

Journal compilation © CSIRO 2018



Label	Description
1	180 mm long × 15-mm diameter stainless steel bolt
2	30-mm stainless steel washers
3	90-mm diameter PVC threaded cap
4	Thread
5	90-mm diameter PVC threaded coupling
6	150 mm long × 75-mm diameter nylon boat roller
7	5-mm hole (drilled)
8	10-mm flare on hole (drilled)
9	180 mm long × 90-mm diameter PVC pipe
10	15-mm lock nut
11	90-mm diameter × 25 mm thick high-density foam
12	10-mm hole through PVC cap and foam

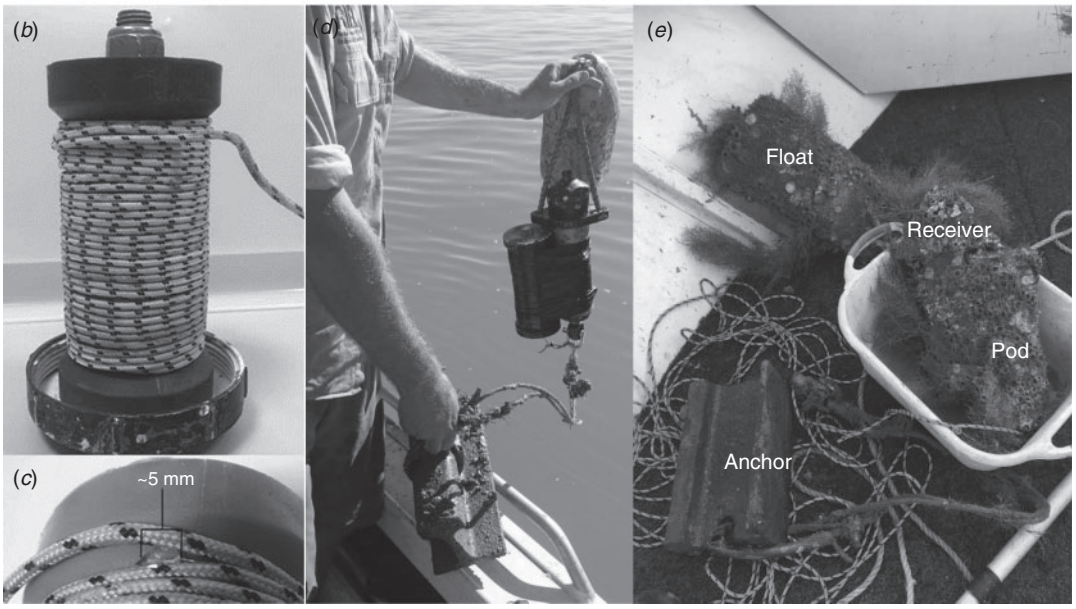
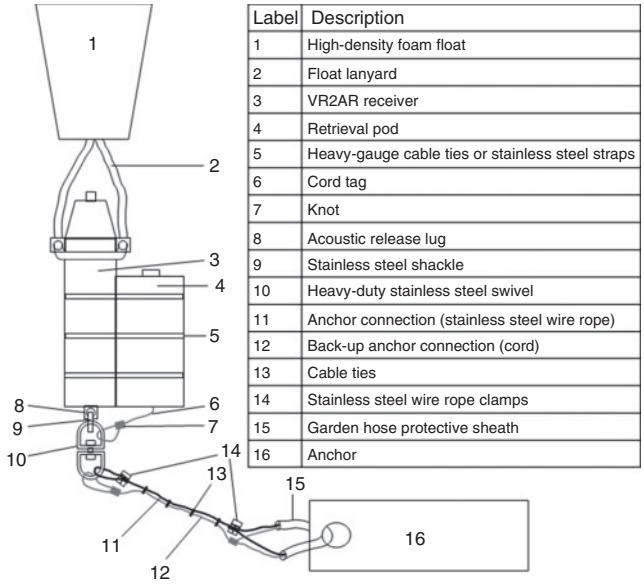


Fig. 1. (*a*) Cross-sectional schematic of a retrieval pod with number labels to identify parts. The cord is not illustrated. (*b*) Nylon roller of a retrieval pod showing 20 m tightly wrapped cord. (*c*) Cord tag fastened to the spool with a small amount of hot glue. (*d*) Deployment of VR2AR receiver (AMIRIX Systems, Vemco Division, Halifax, NS, Canada) and retrieval pod on the Roper River, Northern Territory, Australia. (*e*) Recovered receiver and mooring showing extensive biofouling after 6 months deployment in the Roper River estuary.

mobile transceiver and hydrophone (VR100 and VHTx; https://vemco.com/products/vr100/, accessed 13 September 2017) to disconnect the receiver from its mooring, allowing the receiver to float to the surface where it can be retrieved for data download.

The outer casing of the retrieval pod consisted of a 180 mm-long polyvinyl chloride (PVC) tube (diameter 90 mm) sealed at both ends with PVC threaded couplings and screw-on caps (Fig. 1*a*). Prior to assembly of the tube, 20-m Spectra cord (3 mm, 400-kg breaking strain; Honeywell International Inc., Morris Plains, NJ, USA), was attached to and spooled neatly around a 150 mm-long boat trailer roller (diameter 75 mm) and lightly fixed in place using silicone-based hot glue, leaving an ~300-mm 'tag' (Fig. 1*b*, *c*). A boat trailer roller was used as a spool because the shape allows for neat storage of the cord such that it can unwind rapidly without risk of entanglement. The roller was attached to the upper cap using a 180 mm-long



Label	Description
1	High-density foam float
2	Float lanyard
3	VR2AR receiver
4	Retrieval pod
5	Heavy-gauge cable ties or stainless steel straps
6	Cord tag
7	Knot
8	Acoustic release lug
9	Stainless steel shackle
10	Heavy-duty stainless steel swivel
11	Anchor connection (stainless steel wire rope)
12	Back-up anchor connection (cord)
13	Cable ties
14	Stainless steel wire rope clamps
15	Garden hose protective sheath
16	Anchor

Fig. 2. Schematic of the pod and receiver deployment set-up used in the field study.

stainless steel bolt with lock nut. The top cap was then screwed into place, ensuring enough space around the roller's circumference to allow the cord to run freely from the roller. A 10-mm diameter hole was drilled into the bottom PVC screw cap to allow the cord to pass out of the tube. To reduce incursion of silt and fouling organisms into the tube via this hole, a circle of high-density foam (diameter 90 mm, height 20 mm) was glued into the bottom end of the tube. The cord tag was then passed through a 5-mm diameter hole drilled in the centre of the foam and the lower PVC coupling screwed into place. At the time of writing, the total cost to construct each pod was ~A$70.00. A more detailed description of pod construction and required materials is provided in the Supplementary material to this paper (Table S1).

The pod was then attached directly to the receiver using large plastic cable ties. Both the receiver and pod were wrapped in duct tape and painted with biofouling-preventative paint. The use of duct tape allowed biofouling to be easily removed from the receiver and pod by peeling off the tape after retrieval. The release lug of the receiver was shackled securely to a swivel that was attached to the mooring using plastic-coated stainless steel multifibre rope (Fig. 2). We incorporated a heavy-duty stainless steel swivel into the deployment design to prevent twisting due to underwater currents. Where the wire rope was fastened to the mooring, an additional protective layer (garden hose) was fitted to prevent abrasion against the mooring weight. The cord tag from the pod was attached to the release lug's shackle above the swivel (Fig. 2). We used Spectra cord to add a back-up anchor connection in case the stainless steel wire rope failed. The back-up connection was fixed with cable ties to reduce tangles, but was tied independently to the swivel and anchor (Fig. 2). Once the receiver and pod were secured, they were deployed by carefully dropping the apparatus overboard (Fig. 1*d*).

Following deployment, the acoustic release mechanism was activated using the VR100 transceiver and hydrophone. This disconnected the primary anchor connection (i.e. the release lug) and transferred the buoyancy of the float onto the secondary connection of the retrieval pod. The lift from the float acts to break the hot glue fastening on the cord, allowing the cord to unravel from the roller as the receiver floats to the surface. When the receiver had been brought up onto the boat, the mooring and associated attachments were hauled to the surface using the cord. Given the simplicity of the attachment of the pod to the receiver, it should be straightforward to adapt the pods for use with other types of telemetric or oceanographic equipment.

Field testing

After extensive development and basic deployment testing under controlled laboratory and field conditions, the longer-term reliability of the retrieval pod was tested in a full-scale research study in the estuarine reaches of the Roper River, Northern Territory, Australia. The Roper River estuary is a large, highly erosional system with a tidal range of ~4 m. Six VR2AR listening stations were fitted with retrieval pods and deployed at ~13-km intervals upstream from the estuary mouth in depths ranging from 5 to 8 m. Although reasonably shallow, all sites are considered non-diveable due to the presence of estuarine crocodiles *Crocodylus porosus*, high turbidity and strong tidal current. The receivers were deployed from 18 to 19 May 2016 and retrieved and redeployed on 11–13 October 2016 and again on 31 May–1 July 2017. Each receiver was anchored to the bottom (mud substrate) with a section of railway track weighing 20–30 kg and suspended beneath an ethylene vinyl acetate float (5.7-kg buoyancy; Figs 1*d*, 2).

Of the six retrieval pods activated on each field trip (~6 and 12 months after initial deployment), all were successfully retrieved with no sign of malfunction in the design upon recovery. The anchors of several deployments had sunken into the fine mud substrate. However, given the high breaking strain of the cord (400 kg), we were able to use the power of the boat motor to apply enough force to pull the anchors from the substrate. The outsides of the pods and the receivers had accumulated significant build-up of barnacles and other biofouling, particularly at the more saline downstream sites (Fig. 2*e*). Of the pods with extensive external biofouling, there was only very minor biofouling and accumulation of fine silt on the insides of the pods. Painting the insides of the pods with biofouling-preventative paint is recommended for future deployments to reduce the likelihood of internal biofouling. Despite the very high levels of external biofouling in some instances, there was no sign of any effects of biofouling on the performance of the retrieval system or the structural integrity of the pods.

An important observation made when retrieving receivers during early trials under high tidal flows and strong winds was that the float and receiver may drift rapidly once they reach the surface. If the extent of the cord in the pod is reached before the receiver can be retrieved, the float and receiver may be pulled below the surface, making retrieval more difficult. To prevent this, we recommend filling each retrieval pod with an amount of cord well in excess of the water depth at the deployment site (e.g. >2 : 1 cord-to-depth ratio). In our case, 20 m of cord was used for maximum water depths up to 8 m. If the retrieval pod is being

used in deeper water, it would be necessary to increase the spool capacity to accommodate sufficient cord. If a receiver and its float are dragged under the surface by current or wind during retrieval, it can still be recovered by snagging the cord with a grapple line. It should also be noted that is possible to repeatedly 'range' to the VR2AR using the VR100 and hydrophone if it disappears below the surface.

Conclusion

The retrieval pod described herein enabled the recovery of acoustic telemetry receiver moorings and associated fixings that would otherwise have been discarded, thus reducing the environmental and financial costs associated with the use of the receivers. The components of the retrieval pod are reusable and the pod was successfully field tested in a remote estuarine location, where the receivers and pods were subjected to strong tidal currents, extensive biofouling, occasional extreme weather conditions and high sediment suspension and substrate erosion. Although the conclusions drawn from our trials are limited to 6-month deployments and the habitat under consideration (a large, reasonably shallow estuary), it is likely that the design could be adapted for use in a wide range of aquatic habitats. For example, more cord is required for application in water depths $>$10 m. This could be achieved by increasing the diameter or length of the outer casing of the pod and replacing the standard boat roller with a customised spool with higher carrying capacity. Such modifications would require additional field evaluation to ensure that factors such as increased drag do not interfere with the function of the pod. The retrieval pod system requires minimal construction skills and can be assembled from basic equipment available at most hardware or boating stores, making it a convenient and low-cost option for the retrieval of equipment used with acoustic releases.

Conflicts of interest

The authors declare that they have no conflicts of interest.

Acknowledgements

The authors thank Chris Errity (NT Fisheries) and the Yugul Mangi Rangers, in particular Clarry Rogers and Winston Thompson, at Ngukurr for their assistance with the field work. Peter Kyne (Charles Darwin University), Peter Dostine (NT Department of Environment and Natural Resources) and Hugh Pederson (Vemco) provided useful input on the deployment of receivers, and Matt Northwood (Charles Darwin University) assisted with the construction of the pods. The authors thank Russell Irving and Richard Campbell (Northern Land Council) for facilitating consultations with Traditional Owners. We acknowledge the Traditional Owners of the Roper River region for granting access to the study sites. This research was performed under NLC permit 56390.

References

Fujioka, K., Hobday, A. J., Kawabe, R., Miyashita, K., Itoh, T., and Takao, Y. (2010). Interannual variation in summer habitat use by juvenile southern bluefin tuna (*Thunnus maccoyii*) in southern Western Australia. *Fisheries Oceanography* **19**, 183–195. doi:10.1111/J.1365-2419.2010.00536.X

Heupel, M. R., Semmens, J. M., and Hobday, A. J. (2006). Automated acoustic tracking of aquatic animals: scales, design and deployment of listening station arrays. *Marine and Freshwater Research* **57**, 1–13. doi:10.1071/MF05091

Hussey, N. E., Kessel, S. T., Aarestrup, K., Cooke, S. J., Cowley, P. D., Fisk, A. T., Harcourt, R. G., Holland, K. N., Iverson, S. J., Kocik, J. F., Mills Flemming, J. E., and Whoriskey, F. G. (2015). Aquatic animal telemetry: a panoramic window into the underwater world. *Science* **348**, 1255642. doi:10.1126/SCIENCE.1255642

Huveneers, C., Simpfendorfer, C. A., Kim, S., Semmens, J. M., Hobday, A. J., Pederson, H., Stieglitz, T., Vallee, R., Webber, D., Heupel, M. R., and Peddemors, V. (2016). The influence of environmental parameters on the performance and detection range of acoustic receivers. *Methods in Ecology and Evolution* **7**, 825–835. doi:10.1111/2041-210X.12520

Stark, K. E., Jackson, G. D., and Lyle, J. M. (2005). Tracking arrow squid movements with an automated acoustic telemetry system. *Marine Ecology Progress Series* **299**, 167–177. doi:10.3354/MEPS299167

Webber, D. (2009). VEMCO acoustic telemetry new user guide. Document DOC-004934-01, Amirix Systems Inc., Halifax, NS, Canada.